

Lab 4 (Tutor Copy)**Learning outcomes:**

1. Pre-processing of data for visualisation and analysis
 - a. Able to apply Normalisation to data for comparison on the same axes
2. Present results of analytics visually using an appropriate platform.
3. Interpret the data based on visualisation to find possible relationships between variables in the data

Handling of categorical data type:

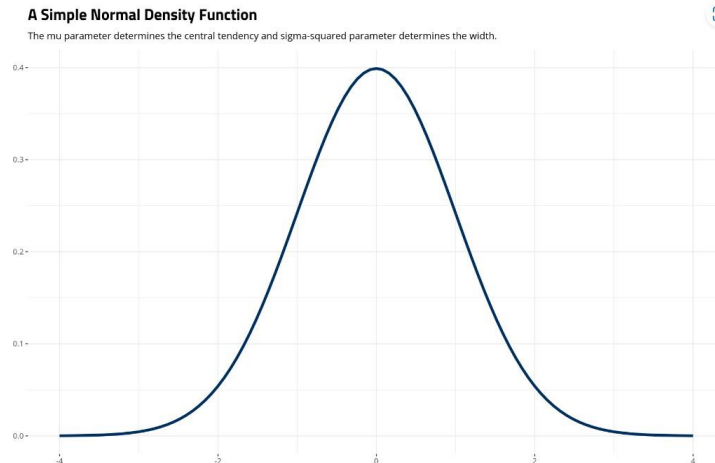
- a. Able to calculate probability
- b. Able to interpret contingency table (cross tabulation)
- c. Able to create pie-chart and bar chart to analyse categorical data types

Activity 1: Understanding Normal and Standard Normal

Often, histogram is presented as density curves. Area under the curves = 1 or 100%.

<https://youtu.be/Txlm4ORI4Gs?si=USCqvFPcNr9sTG0u> (Density Curve explained)

The most common density curve is the **bell-curve**, also known as **normal distribution**.



<http://svmillier.com/images/normal-distribution-central-limit-theorem-inference/super-simple-normal-distribution-1.png>

This type of distribution is “normal” because there is one mode that normally will happen. For example, what is the normal weight of a baby? What is the normal height of a man? Of course, there are small percentage that are higher or lower than normal.

Hence, the bell-curve is a normal distribution. Many statistical hypothesis tests assume the data follow a normal distribution.

List down the characteristics.

1. It is a **bell-shape**.
2. It has **One** mode.
3. The mean and **median** are the same.
4. The area under the curve = **1 or 100%**

Standard Normal Distribution curve

Standard Normal Distribution is just transforming any normal distribution to have the **mean=0** and **standard deviation = 1**

This is the concept that will be used to handle variable that are of different scale or range.

Exercise 1: Compare 2 variables with timeData set: [FictionIceCream_Sale.csv](#)

Question:

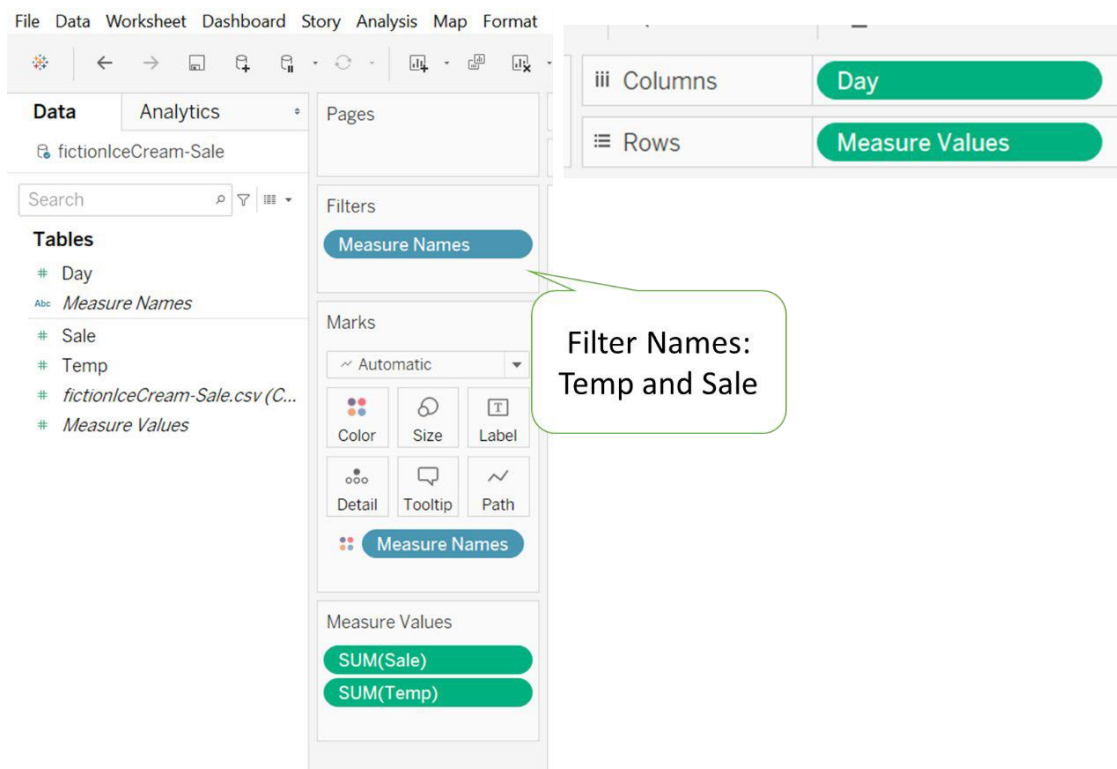
How do Sales of ice-cream and weather Temperature associate with day?

Download the data and copy it to your subject folder “**MyDvis**”

Launch Tableau

Read in the file as “text”.

Start a new sheet. Proceed to plot “Temp” and “Sale” on the same Y-axis (drop to rows), and “Day” on the X-axis (drop to Columns)

[\(Show me how\)](#)

What do you observe? Does the “Sale” increase, when the “Temp” increases?

To have meaningful comparison of 2 variables of different units, we must transform them to the same reference.

Save this workbook. We will return to this exercise after understanding what Normalisation and Feature Scaling is.

Activity 2: Normalisation and Feature Scaling

The **Sales** and **Temperature** values are of different scale.

Sales values being larger has empowered over the “**temperature**”, and we are not able to compare easily.

There are 2 methods to make both variables (Sales & Temperature) of the same range.

Normalisation

$$Z_i = \frac{x_i - \bar{x}}{SD}$$

Converting all individual values (x) to Z.

Feature scaling

$$X' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

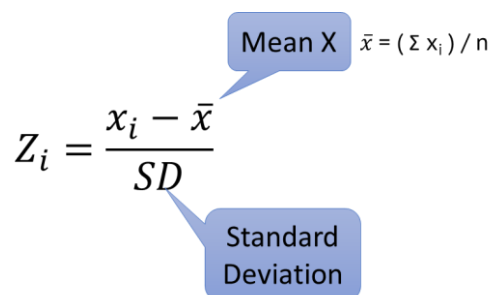
Converting all individual values(x) to X'.

The transformation makes the final Z-score to have mean = 0 and Standard Deviation = 1. This is applying the Standard Normal curve.

Feature Scaling is a scaling method, to map the minimum to 0 and maximum value to 1.

This result in both variables will have a range from 0 to 1.

How to apply Normalisation?

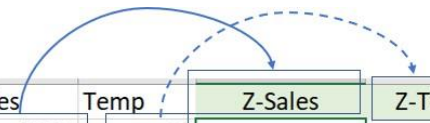


The diagram shows the formula $Z_i = \frac{x_i - \bar{x}}{SD}$. A callout box labeled "Mean X" points to $\bar{x}$ with the formula $\bar{x} = (\sum x_i) / n$. Another callout box labeled "Standard Deviation" points to SD .

Apply this formula to every x-value.

So, for every Sales, calculate the corresponding Z-sale.

For every Temp, calculate the corresponding Z-Temp.



Day	Sales	Temp	Z-Sales	Z-Temp
1	1000	26		
2	1200	28		
3	1800	30		
4	2500	31		
5	1700	28		
6	1700	27		
7	1050	25		
8	1100	27		
9	1700	31		
10	2600	32		
11	1500	27		
12	1800	26		
13	1000	26		
14	1200	28		
15	1800	30		

To calculate the corresponding Z-score, we need to know the Mean And Standard Deviation of the original set.

	Sales	Temp
mean	1576.67	28.1333
SD	503.866	2.16685

Instead of manually converting each temp and sales value to Z-temp and Z-sale, we will write a formula for the program to compute.

In TABLEAU, the Calculated Field is used to input the formula.

Continue with previous exercise on [fictionIceCream_sales.csv](#) data.

Exercise 2: Create Calculated Field in TABLEAU for Normalisation

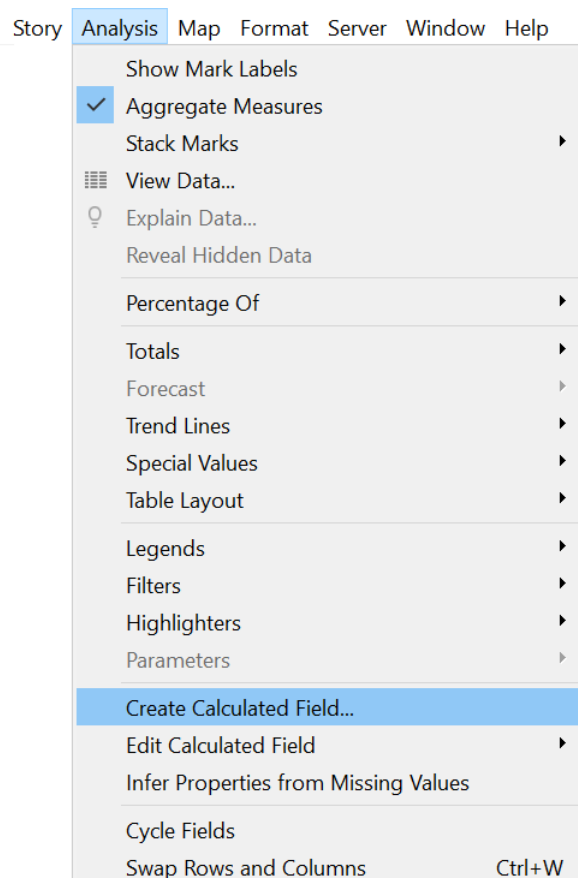
We will create two new fields; one to transform the “Temperature” and another to transform the “Sale”.

One named “Z-Temp”, which is the new Z-score of the “Temperature”.

Another named “Z-Sale” which is the new Z-score of the “Sale”.

[\(show me how\)](#)

Create a New Calculated Field.



Enter the following formula for “Z-Temp”:

$([Temp] - \{ FIXED : AVG([Temp]) \}) / \{ FIXED : STDEV([Temp]) \}$

$$Z_i = \frac{x_i - \bar{x}}{SD}$$

×

$$\frac{([Temp] - \{ FIXED : AVG([Temp]) \})}{\{ FIXED : STDEV([Temp]) \}}$$

The calculation is valid.
Apply
OK

Click “OK”. Under the Table tab, you will notice the newly created variable with the icon =#.

Tables

#	Day
Abc	Measure Names
#	Sale
#	Temp
=#	Z-Temp
#	fictionIceCream-Sale.csv ...
#	Measure Values

Create a new calculated field for “Z-Sales”:

Enter the following formula for “Z-Sales”:

$$\frac{([Sales] - \{ FIXED : AVG([Sales]) \})}{\{ FIXED : STDEV([Sales]) \}}$$

Z-Sales



$$([Sales] - \{ FIXED : AVG([Sales]) \}) / \{ FIXED : STDEV([Sales]) \}$$

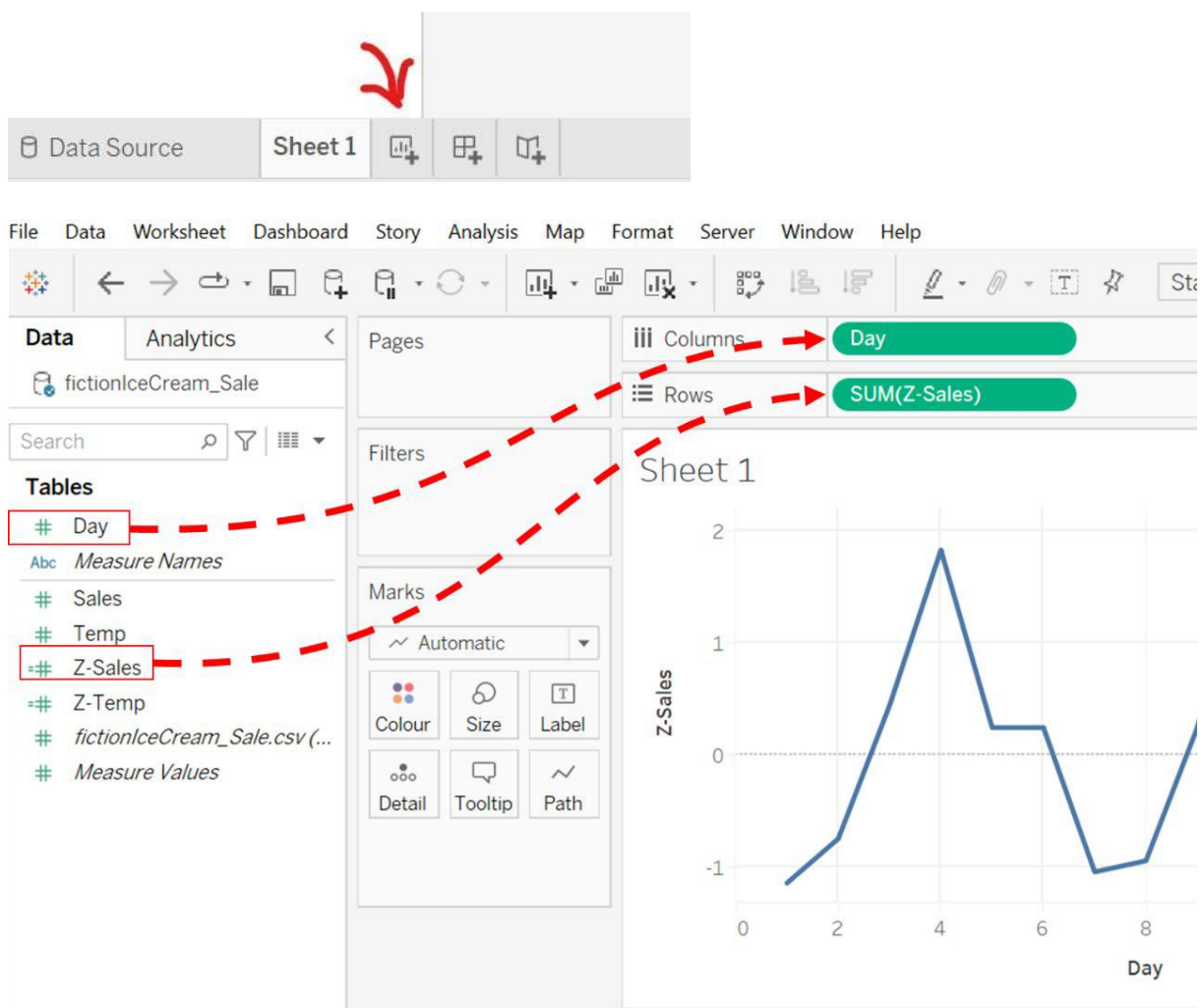
The calculation is valid.

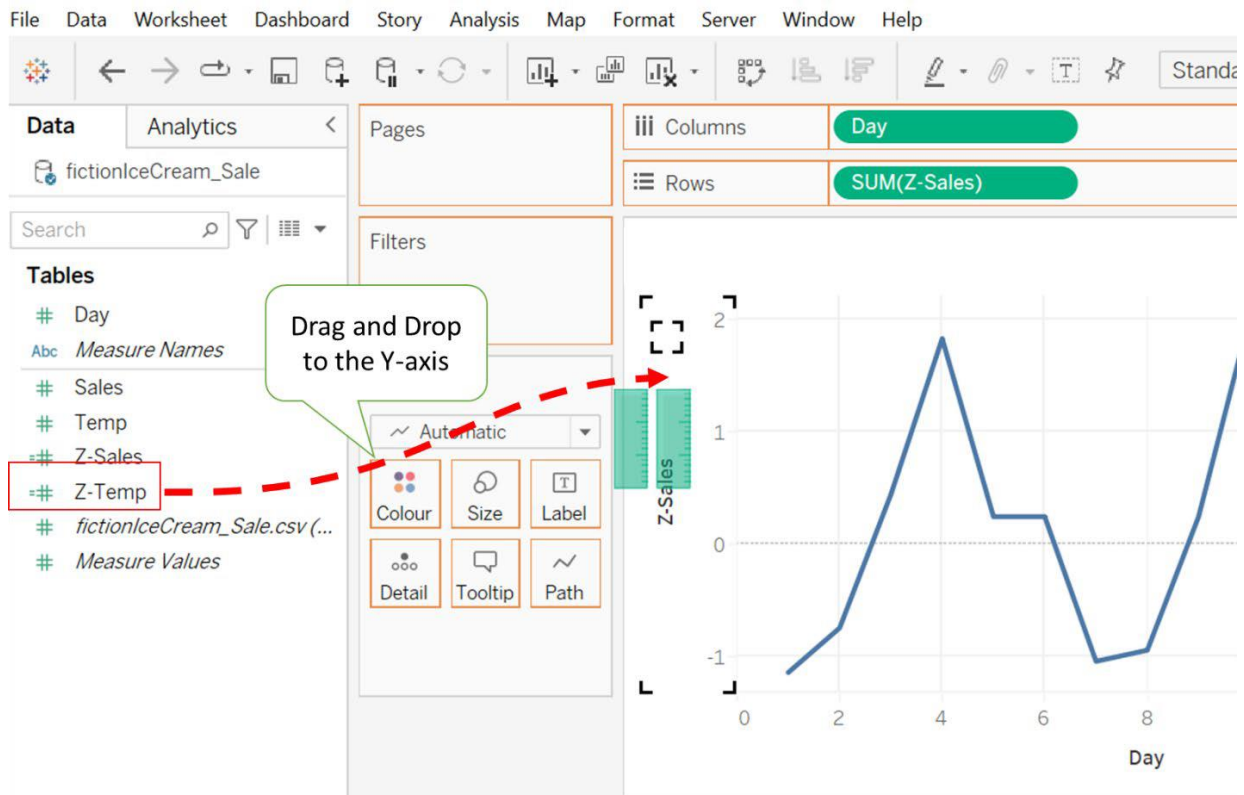
Apply

OK

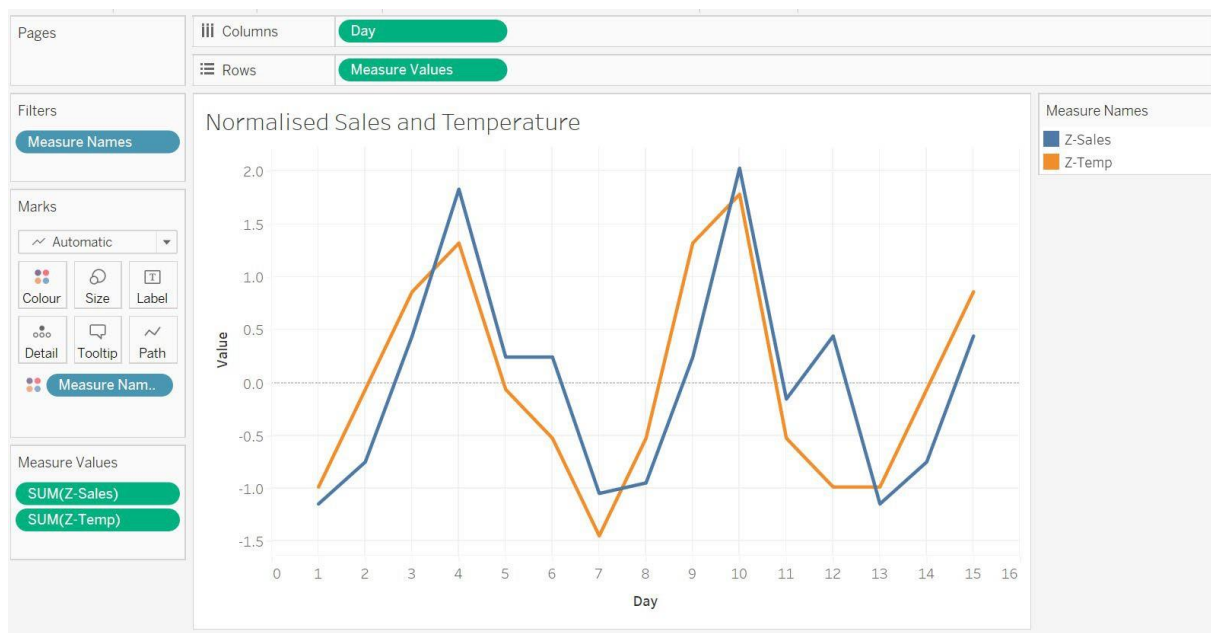
Click "OK" to continue.

Create a new Sheet.





What do you observe?



Both Z-Sales and Z-Temp now have Mean =0 and Standard Deviation of 1.

Note: The Y-axis shows the normalised range, not the actual measured value of Sales and Temperature.

Observation/insights (tutor can share with the students):

*Day 4 and Day 10 show an increase in temperature. The ice-cream sales also increase.
Day 7 and Day 13 show a decrease in temperature, and the ice-cream sales also dropped.
There is a strong correlation between the temperature of the day and the ice-cream sales.*

Manual Calculation Practice

For online test, you will perform manual calculation to determine the Z-score.

Example:

The following are the test results of 5 students.

60, 75, 80, 90, 95

The Mean is 80, and sample standard deviation is 12.25.

Calculate the Z-score for every marks.

Marks (X)	Z-score
60	$(60 - 80)/12.25 = -1.6326$
75	$(75 - 80)/12.25 = -0.4082$
80	$(80 - 80)/12.25 = 0$
90	$(90 - 80)/12.25 = 0.81632$
95	$(95 - 80)/12.25 = 1.2245$

Exploratory Data Analysis

We have covered a few ways to explore numerical data:

- Finding the Central Tendency and Spread via Statistics, box plot and histogram.
- Finding the linear correlation among the variables via scatter plots and linear correlation matrix
- Transform the variables so as to compare them in the same timeline

These methods will help us to gather preliminary insights.

With more variables, we can drill further into comparing two or more variables for more insights.

But all these methods can only use for numerical data types.

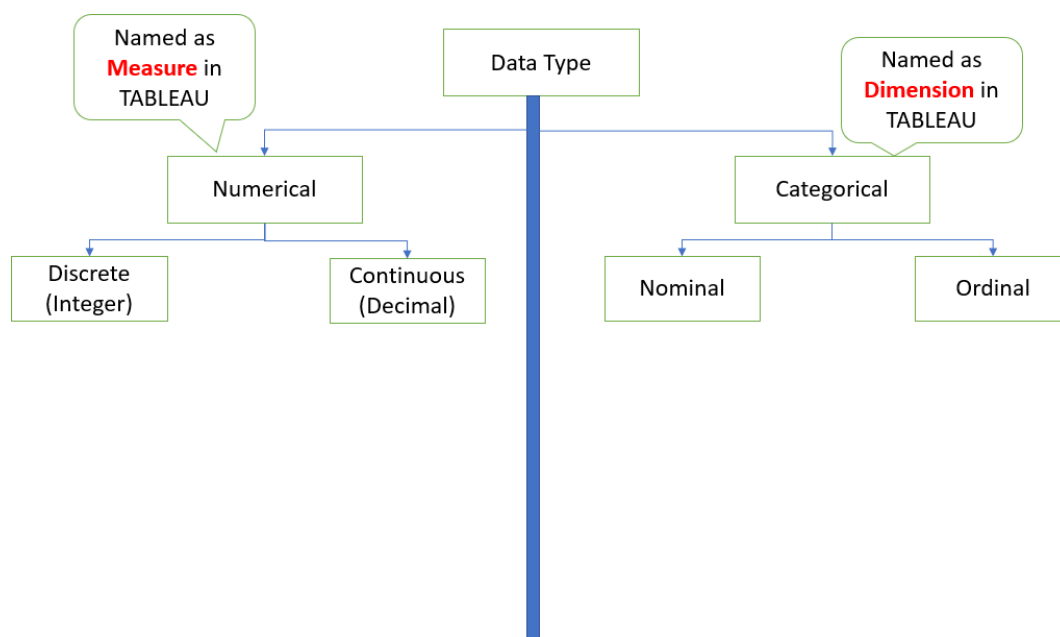
There are data that is categorical, also known as “nominal”.

They are not “numbers” and their values are not meant to be “sum”.

For example, Gender, types of services, high or low states (digital output gathered from IoT sensors), Agree or Disagree, countries, present or absent, etc...

Categorical or Nominal data type are mostly “STRING” (in computer terms). Methods to explore such data will be count of occurrences, expressed in probability percentage, conditional probability (chances of happening).

In the next exercises, we will recap the topic on probability.



Activity 3: Probability Concept

(a) What is the probability of throwing a dice to get even number showing up?



Step 1: Find out the number of all possible outcomes

All Possible outcome = { 1, 2, 3, 4, 5, 6}

Number of all possible outcomes = 6

Step 2: Find out the number possible favorable outcomes

Set of even number = {2,4,6}

Number of favourable outcomes = 3

$$P(\{2,4,6\}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

$$\text{Probability of even numbers} = \frac{3}{6} = \frac{1}{2}$$

(b) Suppose a coin is flipped 3 times. What is the probability of getting two tails and one head?



Total Possible outcomes

S = {TTT, TTH, THT, THH, HTT, HTH, HHT, HHH}

Number of possible outcomes = 8

Favourable outcomes

2 Tails and 1 Head combination = {TTH, THT, HTT}

Number of favourable outcomes = 3

Probability of getting 2 tails and one head = 3 out of 8 = 3/8

(c) In a sample of 43 students:

15 had brown hair.

10 had black hair.

16 had blond hair.

2 had red hair.

Fill in the frequency distribution table.

<i>Hair Color</i>	<i>Frequency</i>
Brown	15
Black	10
Blond	16
Red	2

Number of students having hair color not red nor blonde?

Number of students with red or blond hair = $2 + 16 = 18$

What is the probability of a student having hair color not red nor blond?

$P(\text{Red or blond}) = 18/43$

$P(\text{not Red or Blond}) = 1 - P(\text{Red or Blond})$

$= 1 - (18/43)$

$= 25/43$

(d)

Gender \ Handed-ness	Handed-ness		Total
	Right handed	Left handed	
Male	43	9	52
Female	44	4	48
Total	87	13	100

This is a contingency table, also known as CrossTabs.

The first column is the possible outcomes for variable 1, which is “Gender’ and first row is the possible outcomes for variable 2, which is “Handed-ness”.

Total number of people surveyed: **100**

Total number of Male in the survey: **52**

Probability of a Male who is left-handed : **9/100**

Percentage of female **among** the left-handed: **4/13**

(e)

		Medicine Taken		
Cold Length		yes	no	Total
	1 -3 days	86	19	105
	4 - 7 days	16	79	95
	Total	102	98	200

1. What is the probability to recover in 1-3 days?
 $105/200 = 0.525$ or **52.5%**
2. What is the probability of taking the medicine and recover in 1-3 days?
 $86/200 = 0.43$ or **43%**

(f)

A medical trial into the effectiveness of a new medication was carried out. 120 females and 90 males took part in the trial. Out of those people, 50 females and 30 males responded positively to the medication.

Form the Contingency Table.

	Female	Male	Total
Positive	50	30	80
Negative	70	60	130
Total	120	90	210

Total Sample Numbers

1. What is the probability that the medicine shows a positive result among the females?

$$50/120 = 5/12 \text{ or } 41.7\%$$

2. What is the probability that the medicine shows a positive result among the males?

$$30/90 = 1/3 \text{ or } 33.3\%$$

3. What is the probability of finding a participant who is a female and shows positive result?

$$50/210 = 5/21 \text{ or } 23.8\%$$

Exercise 3: Visualisation for categorical data

Data Set: [mental_health_small.csv](#)

This was a survey returns of IT Professionals from various countries on their views of their employer's attitudes towards and support for the mental health of employees. For more details (e.g., the survey questions for each column), you can visit this link (<https://www.kaggle.com/osmi/mental-health-in-tech-survey/data>)

Refer to Appendix (next 2 pages) for the survey questions and the variables that stored the answer to the question.

Appendix

Data File: [mental_health_small.csv](#)

This dataset contains the following data field:

Variable	Survey questions
Timestamp	(Date & Time that the survey was submitted)
Age	Enter your age
Gender	M or F
Country	Name of the country
State	If you live in the United States, which state or territory do you live in?
self_employed	Are you self-employed?
family_history	Do you have a family history of mental illness?
treatment:	Have you sought treatment for a mental health condition?
work_interfere	If you have a mental health condition, do you feel that it interferes with your work?
no_employees	How many employees does your company or organization have?
remote_work	Do you work remotely (outside of an office) at least 50% of the time?
tech_company	Is your employer primarily a tech company/organization?
benefits	Does your employer provide mental health benefits?
care_options	Do you know the options for mental health care your employer provides?
wellness_program	Has your employer ever discussed mental health as part of an employee wellness program?
seek_help	Does your employer provide resources to learn more about mental health issues and how to seek help?

anonymity	Is your anonymity protected if you choose to take advantage of mental health or substance abuse treatment resources?
Leave	How easy is it for you to take medical leave for a mental health condition?
mental health consequence	Do you think that discussing a mental health issue with your employer would have negative consequences?
phys health consequence	Do you think that discussing a physical health issue with your employer would have negative consequences?
coworkers	Would you be willing to discuss a mental health issue with your coworkers?
supervisor	Would you be willing to discuss a mental health issue with your direct supervisor(s)?
mental health interview	Would you bring up a mental health issue with a potential employer in an interview?
phys health interview	Would you bring up a physical health issue with a potential employer in an interview?
mental vs physical	Do you feel that your employer takes mental health as seriously as physical health?
obs_consequence	Have you heard of or observed negative consequences for coworkers with mental health conditions in your workplace?
comments	Any additional notes or comments

Number of observations: **982**

Number of variables (excluding date&time): **24**

Which variables are numerical data type? **Age**

Notice this survey consists of data that are mostly categorical. The values are either “yes” or “no” or “maybe”.

There is no statistics for categorical data type.

Most common visual chart is Bar Chart and Pie Chart.

How to create a Pie-chart in Tableau:

Preliminary question:

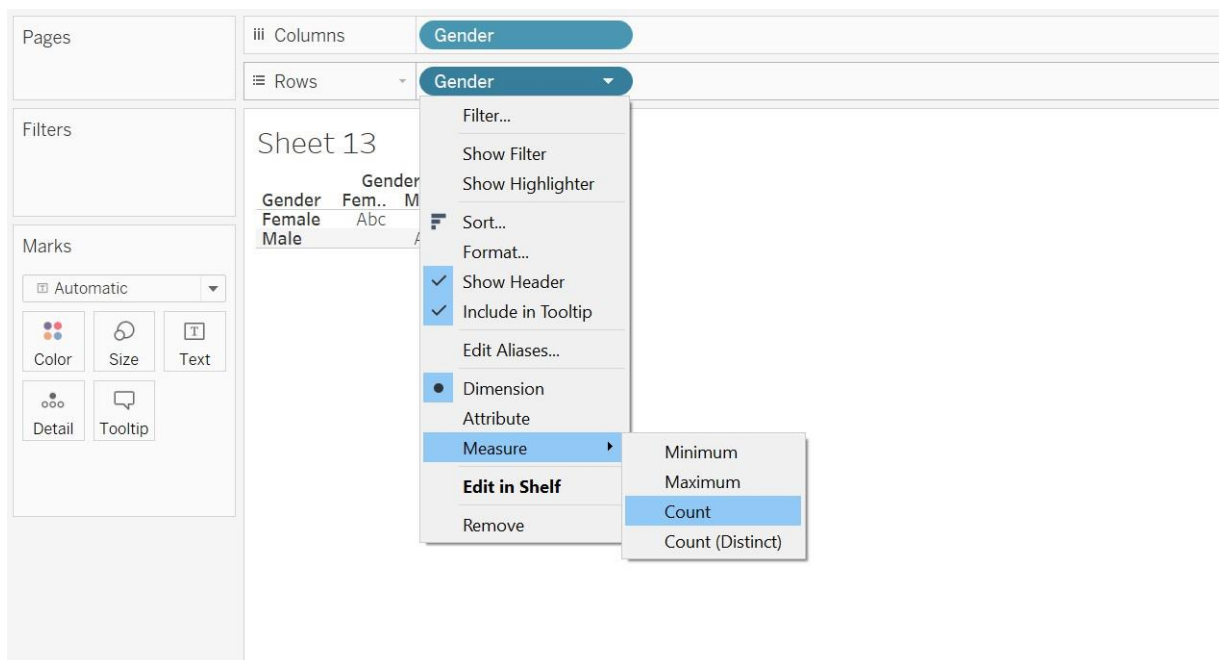
1. What is the distribution of males and females in the survey?

[\(Show me how\)](#)

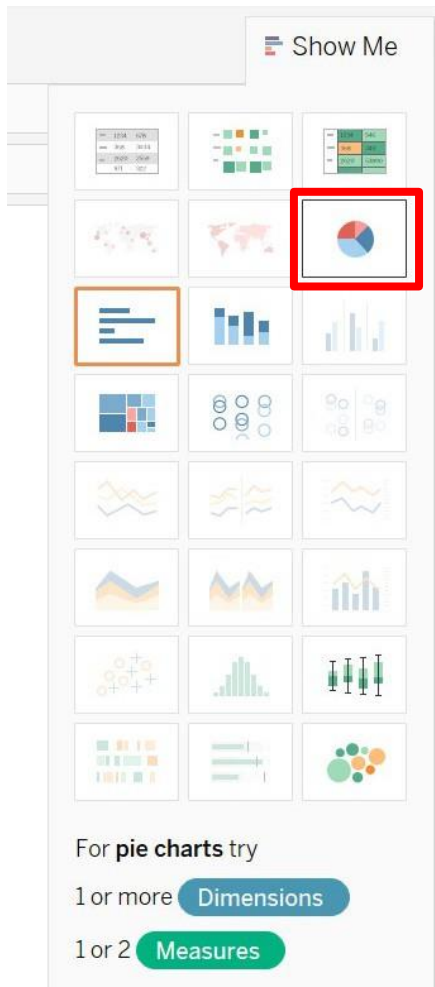
First, is to create a bar chart:

Drag the variable “Gender” to both columns and rows.

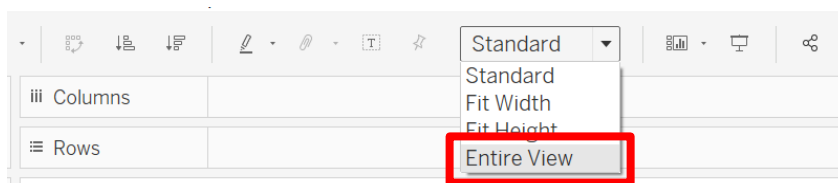
Edit the option for the “Gender” in Rows as “Count”.



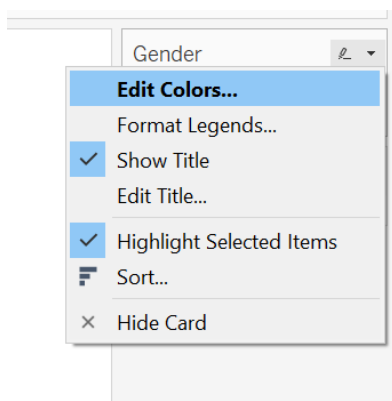
Then, open-up the “Show Me” tab (on the right), and select “pie-chart”:



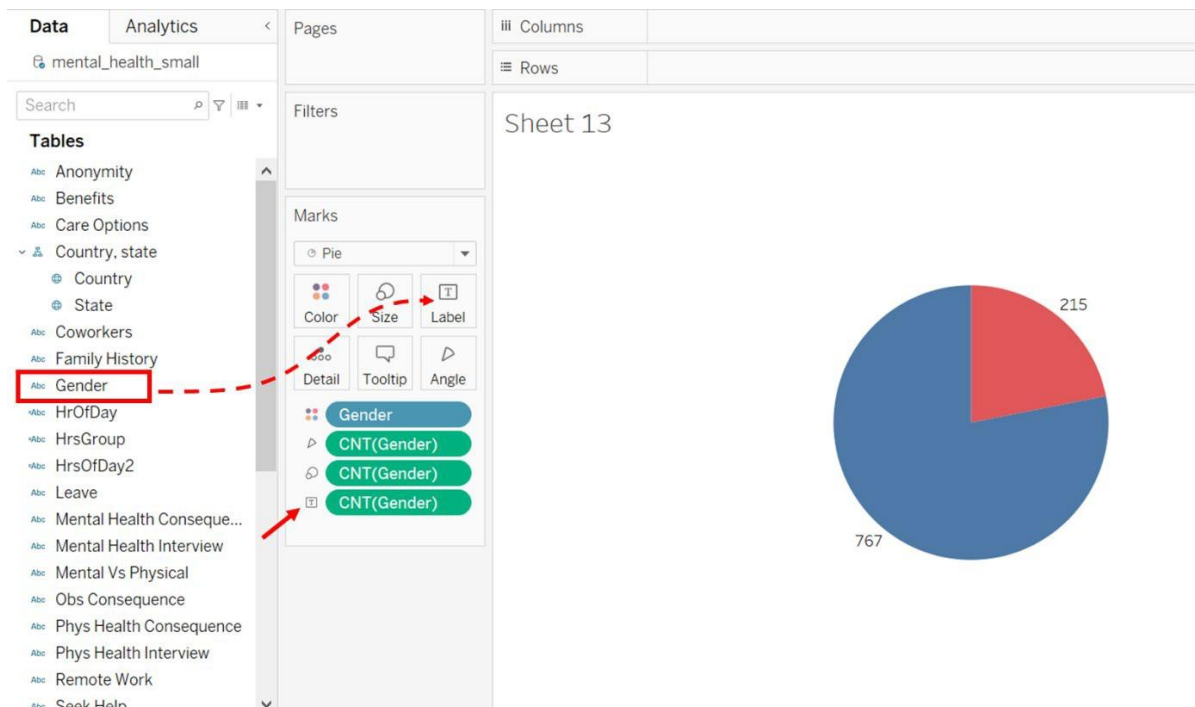
Extend the chart to full view:



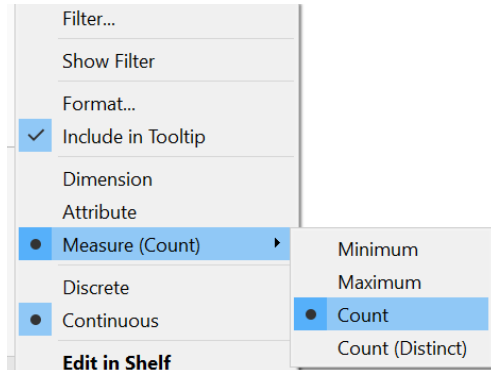
You change the legend color from:



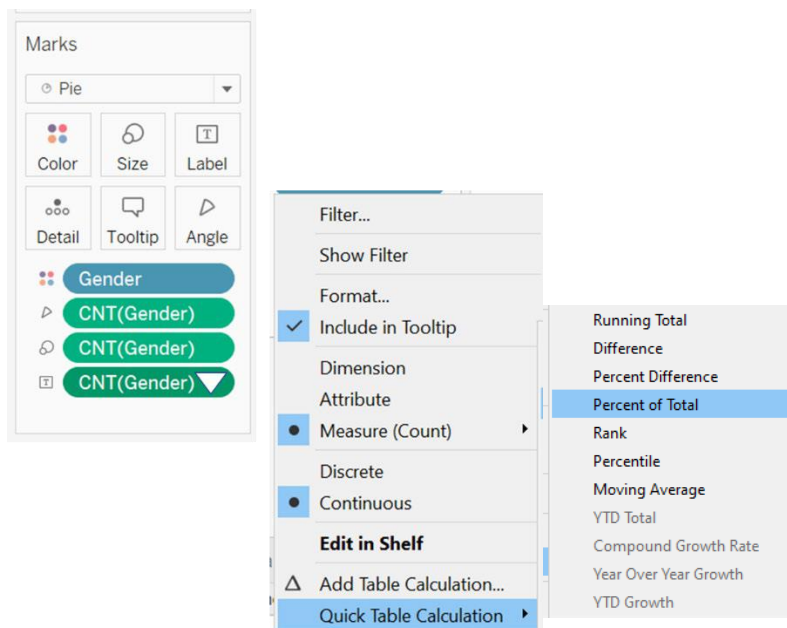
To display the numbers on the chart: Drag “Gender” to Label marks.



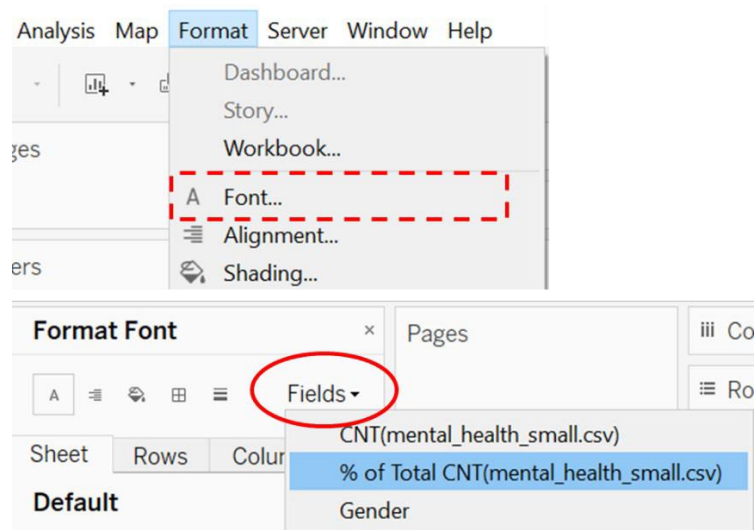
Change the measure to “count”.



To show as Percentage:



To edit the number of decimal places:



Format % of Total CNT(...)

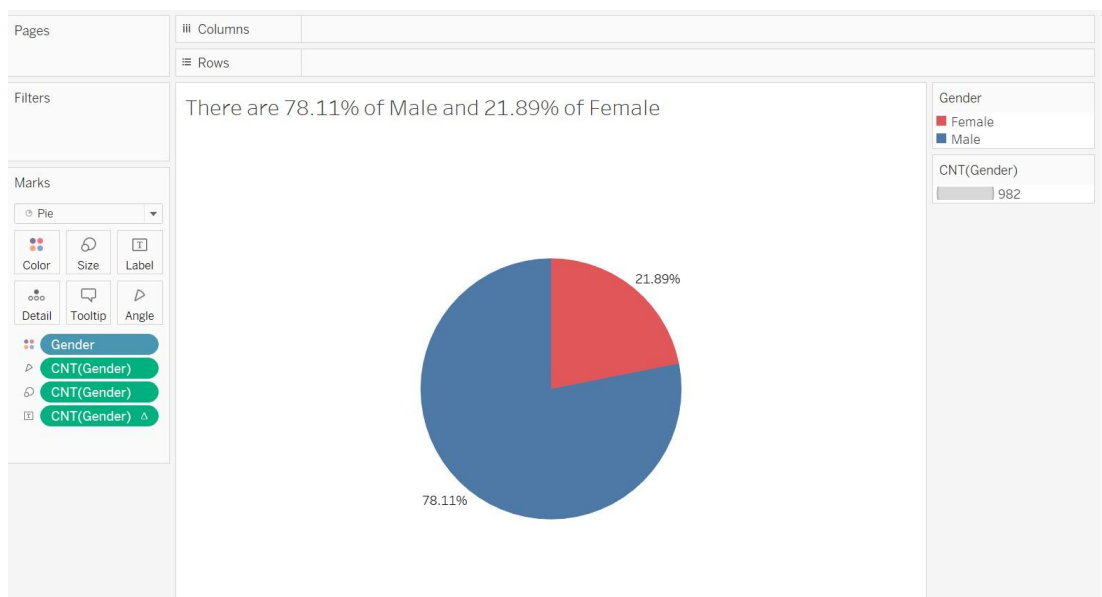
Default
Font: Tableau Bo..
Alignment: Automatic
Numbers: 12345600..

Totals
Font: ..
Alignment: ..
Numbers: ..

Grand Total:
Font: ..

Percentage
Decimal places: 2

Click (X) to return to the data pane.



2. Are there more male or female with family history of mental illness?

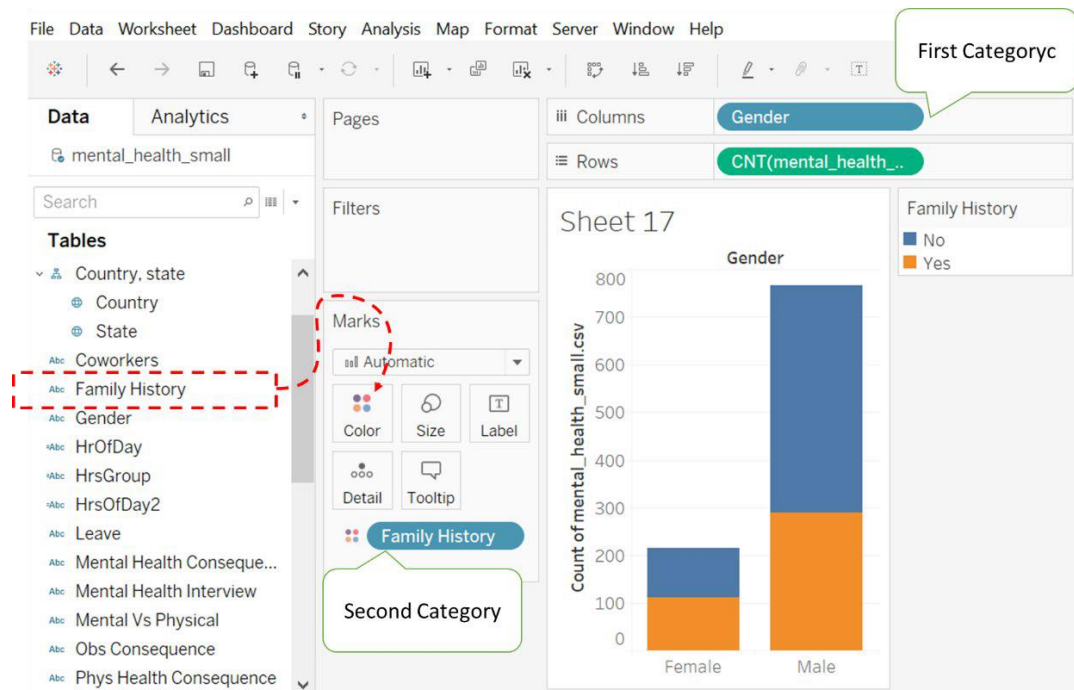
This question involves variable **“Gender”** and **“Family history”**.

Start with a bar chart for one variable “Gender”.

How to create stack bar-chart?

Stack-bar is used to show 2 categories on the same chart.

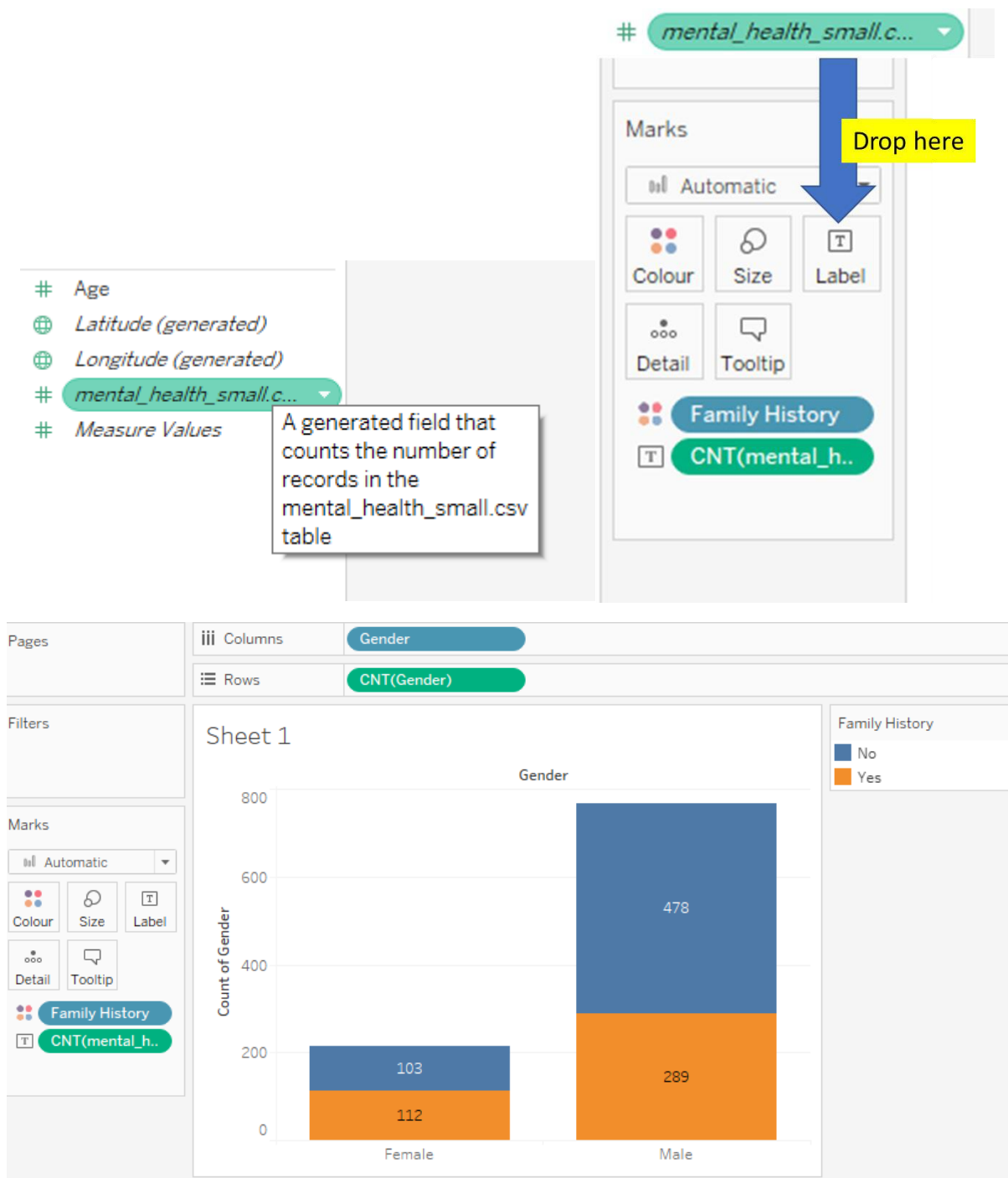
This is making use of the “color” marks to represent variable “Family history”.



This chart shows the distribution of Gender (x-axis). The “Color” represents the category “Family History” which consist of values “yes/no”.

How can we compare, which gender has more number of “family history”?

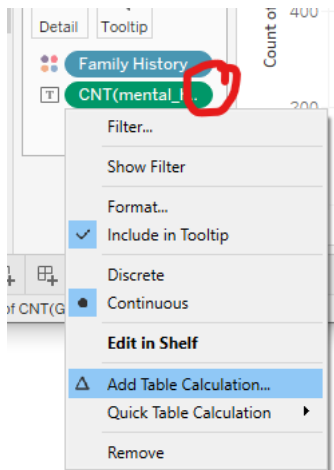
We can drag the “count” generated by TABLEAU to the “label” marks:



Is it fair to say male has more with family history than female?

We should find the % number of “family history” among each gender group.

Instead of “count”, edit to “add a table calculation”:

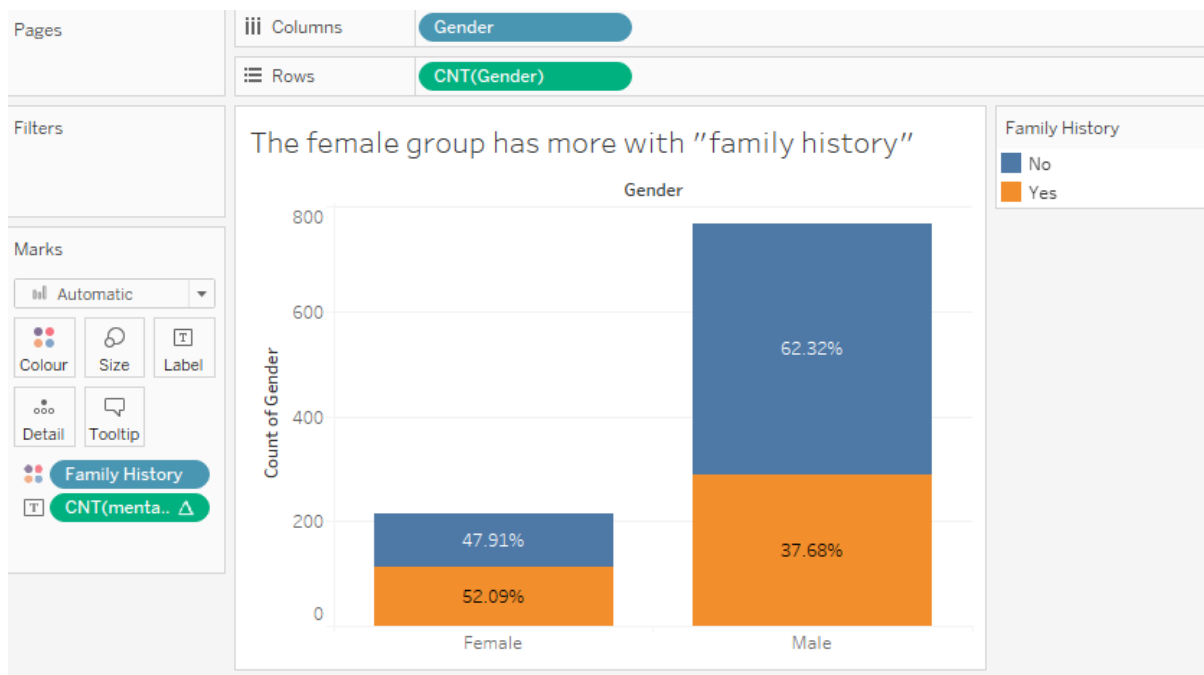


(click on the down arrow and select “Add Table Calculation”)

Dialog box below pop-up.

A screenshot of the 'Table Calculation' dialog box. The title bar says 'Table Calculation' and the subtitle is '% of Total Count of mental_health_small.csv'. There is a close button (X) in the top right corner. The 'Calculation Type' section has a dropdown menu set to 'Percent of Total'. Below it is a checkbox for 'Compute total across all pages'. The 'Compute Using' section has a dropdown menu with options: Table (across), Table (down) (selected and highlighted with a blue arrow), Table, Cell, and Specific Dimensions. To the right of the 'Table (down)' option is the text 'To compute column total'. Below the dropdown are two checkboxes: 'Family History' (checked) and 'Gender' (unchecked). At the bottom, there is a dropdown for 'At the level' and a checked checkbox for 'Show calculation assistance'.

The update is instance on the chart.
Click on the “cross” symbol on the top right corner to close this dialog box.

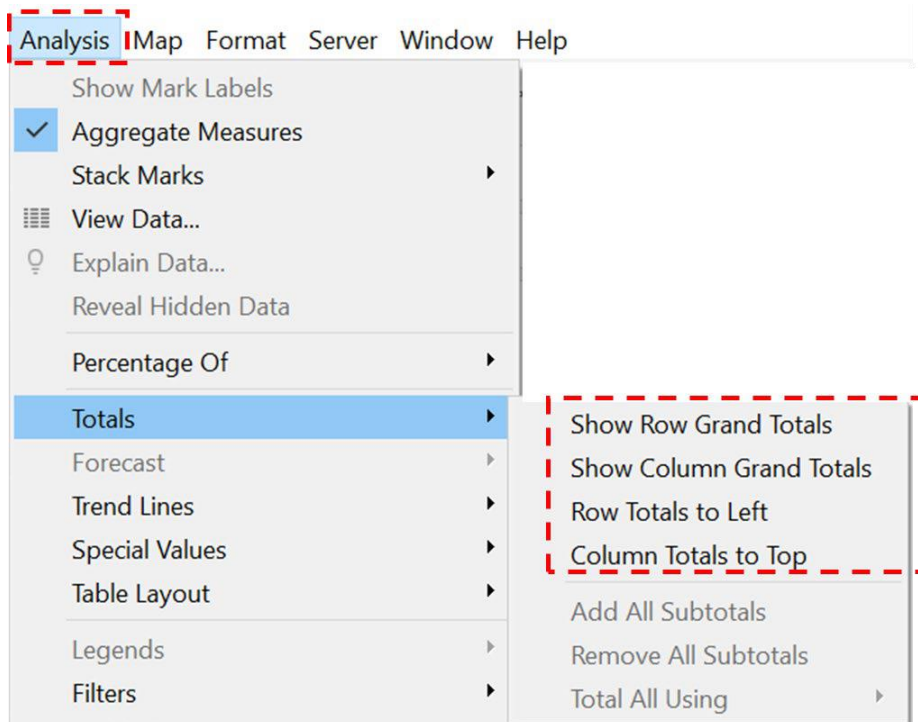


Tips:

Stack bar can be easily displayed into contingency table with the help of "Show me":



You can compute the total for each rows and columns, as well as grand total from "Analysis" menu.



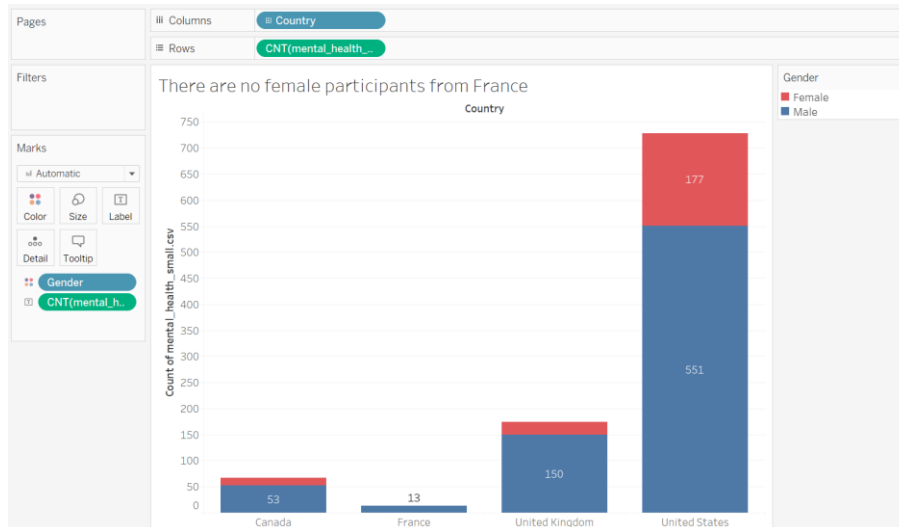
This will give the Cross Tabs:

Gender	Family History		Grand Total
	No	Yes	
Female	103	112	215
Male	478	289	767
Grand ..	581	401	982

Ask more questions:

3. What is the breakdown of survey respondents by country and gender?

*Use Stack bar-chart

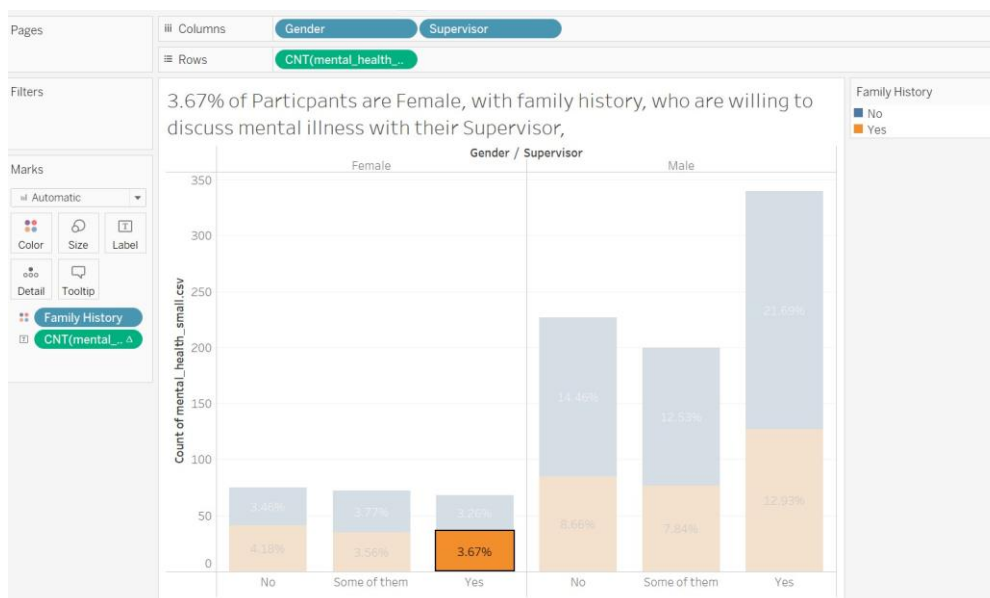


Most of the participants are from United States. The least number of participants are from France.

[\(Show me how\)](#)

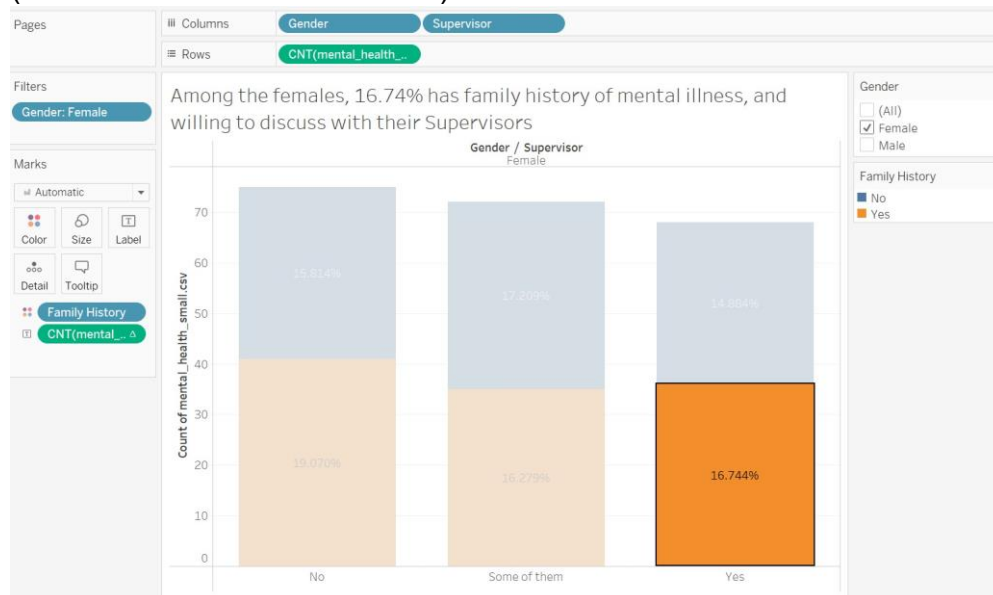
4. What is the percentage of IT professional, who are females (C), and willing to discuss mental issues with their Supervisors (A) **and** have family history of mental illness (B)? $P(A \cap B \cap C)$

(ans: $36/982 = 0.0367$ or 3.67%)



[\(Show me how\)](#)

5. **Among** the female IT professionals (C), what is the percentage that they are willing to discuss mental issues with their Supervisors (A), and have family history of mental illness (B)? $P(A|B|C)$
(ans: $36/215 = 0.167$ or 16.7%)



[\(Show me how\)](#)

6. **Ask 2 further questions and present your insights. (Submit to LMS)**

Additional Practices:

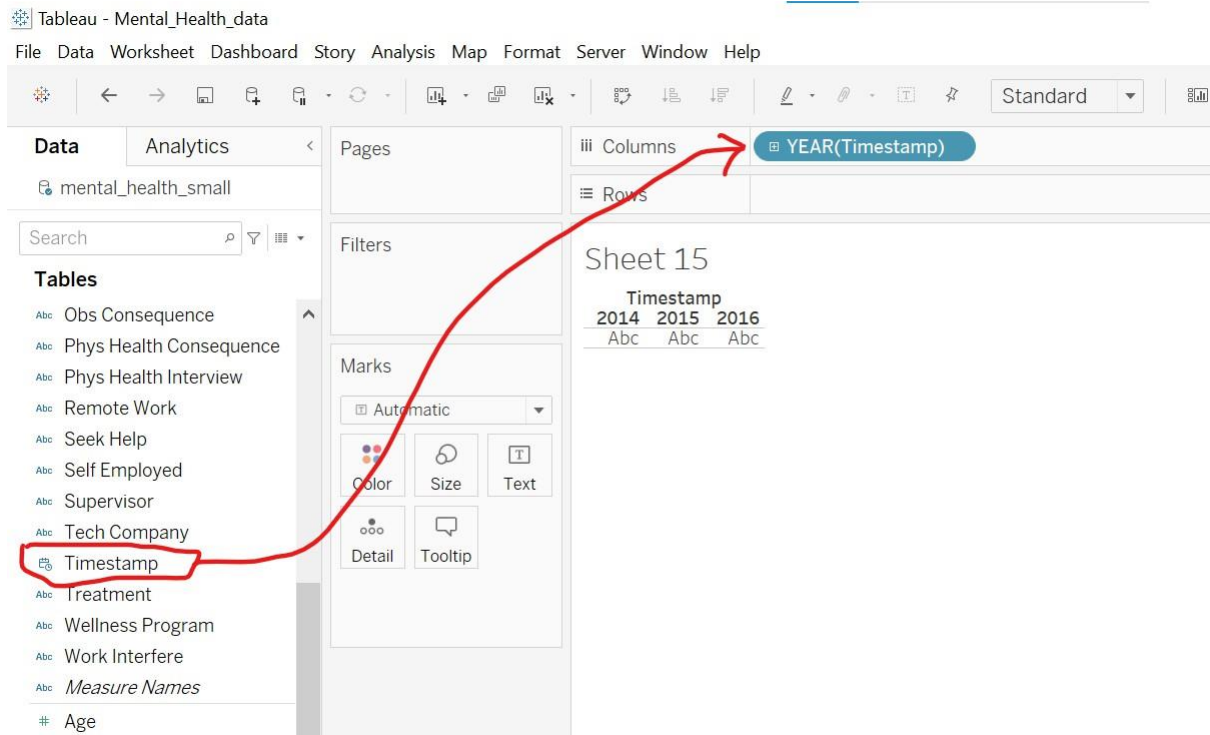
Most participants took the survey at which time of the day?

Plotting variable Vs timeline

The timestamp shows the date and time that participant submitted his/her survey.

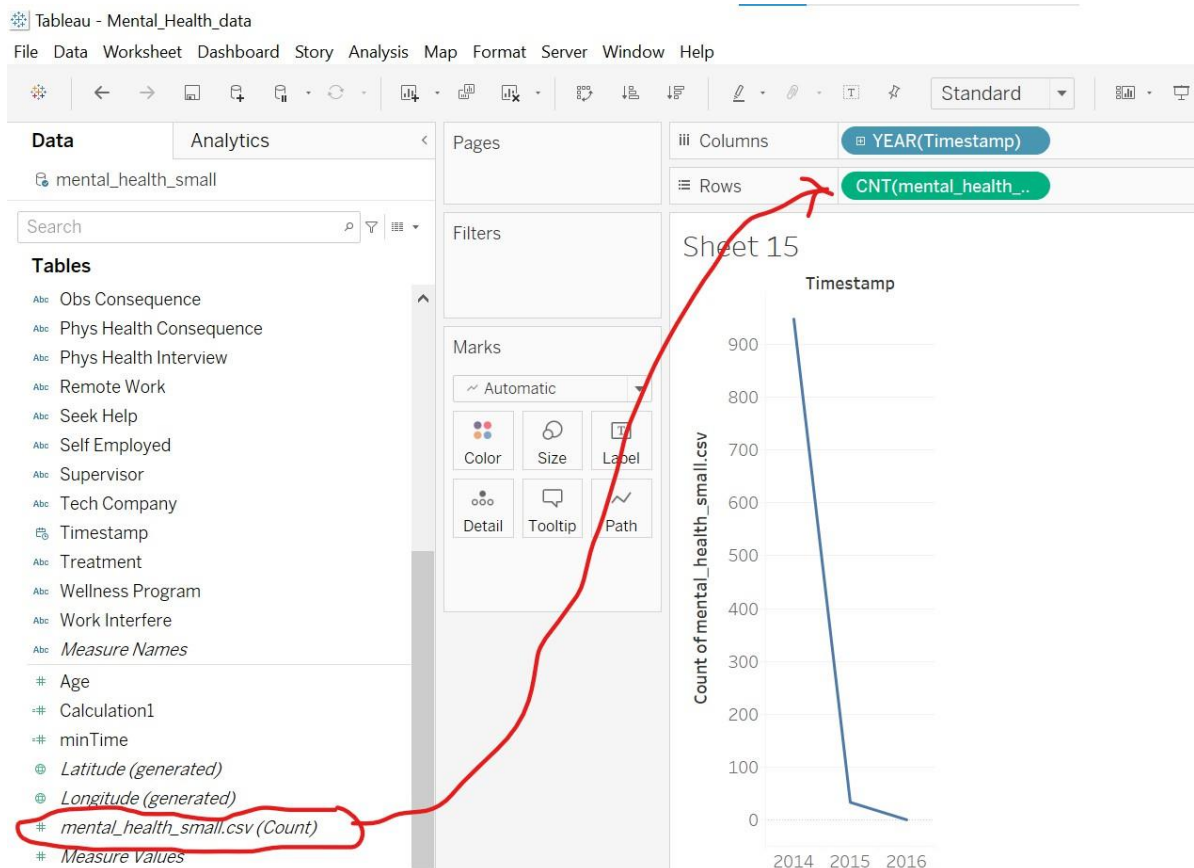
Sort fields		Data source order	
mental_health_small.csv		#	Abc
Timestamp		Age	Gender
27/8/2014 11:29:00 AM		37	Female
27/8/2014 11:29:00 AM		44	Male
27/8/2014 11:29:00 AM		32	Male
27/8/2014 11:29:00 AM		31	Male
27/8/2014 11:30:00 AM		31	Male
27/8/2014 11:31:00 AM		33	Male
27/8/2014 11:31:00 AM		35	Female

We are interested to know which time of the day when most participants submit their survey. The time that we are interested will be which hour of the day (instead of minutes or seconds). We can drag the Timestamp to the “column”, which will be the label for the X-axis.



The table shows there are 3 years, 2014, 2015, 2016.

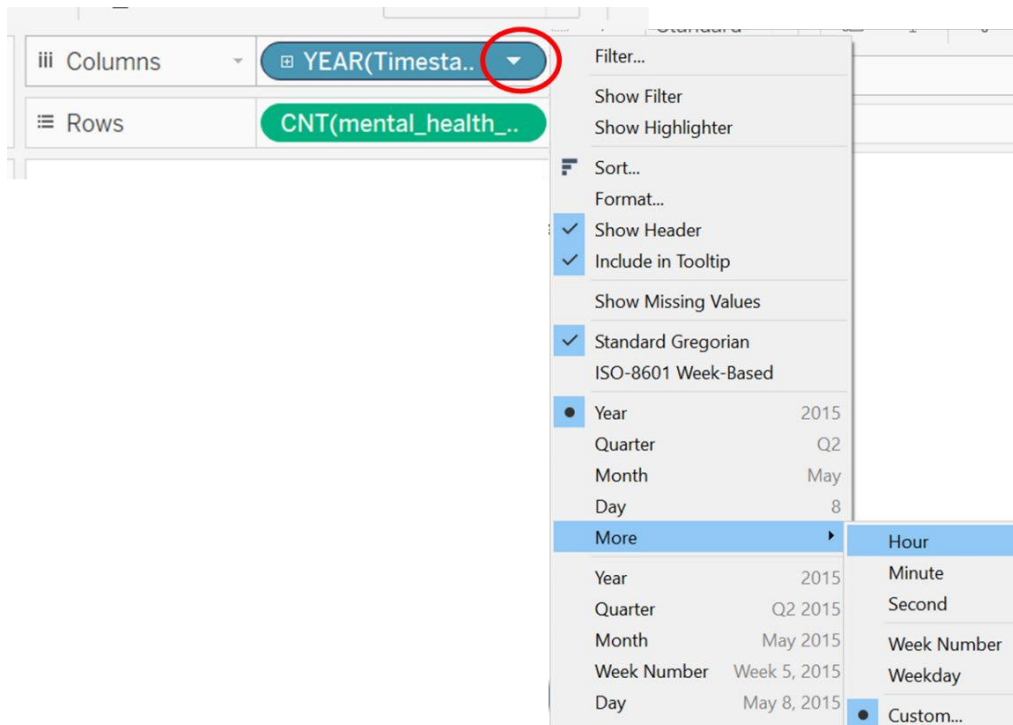
We can drag the generated count field to the “row”:



The X-axis shows the 3 years mark, and the Y-axis shows the count from each year.
2014 has the most submissions.

We are interested in which time of the day has the most submissions. The time units will be in hour (not day, minutes nor seconds).

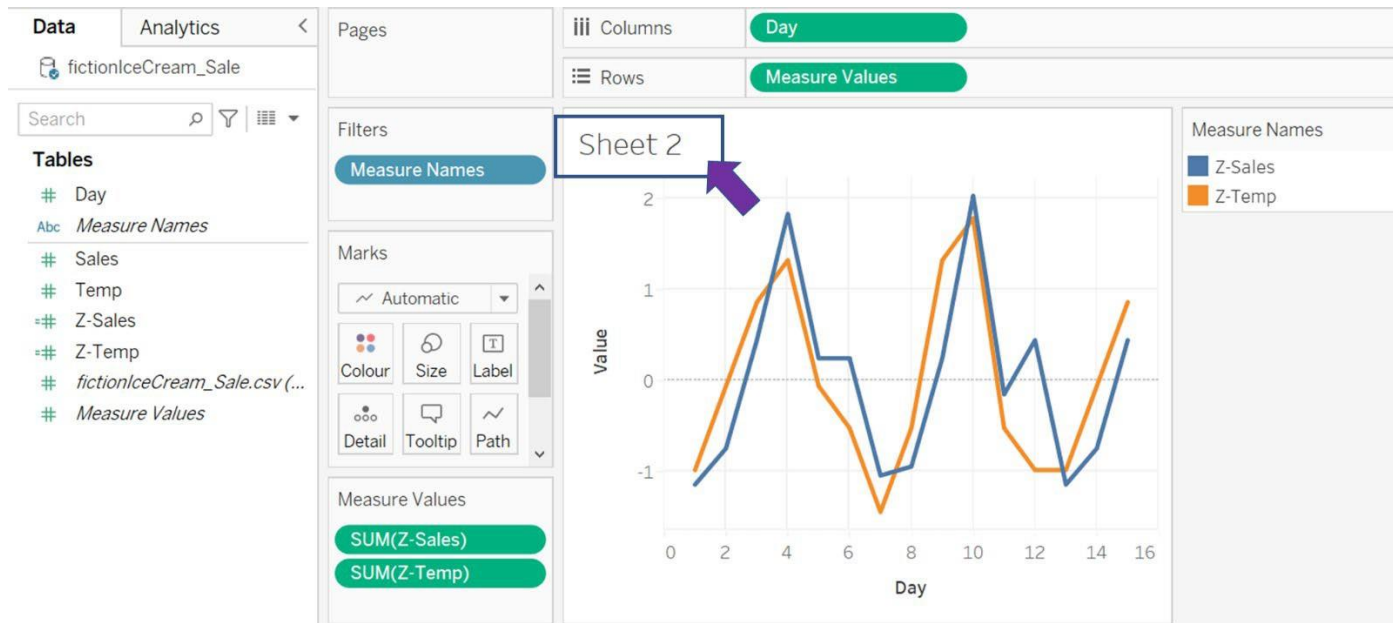
Edit the “timestamp” to “hour”:



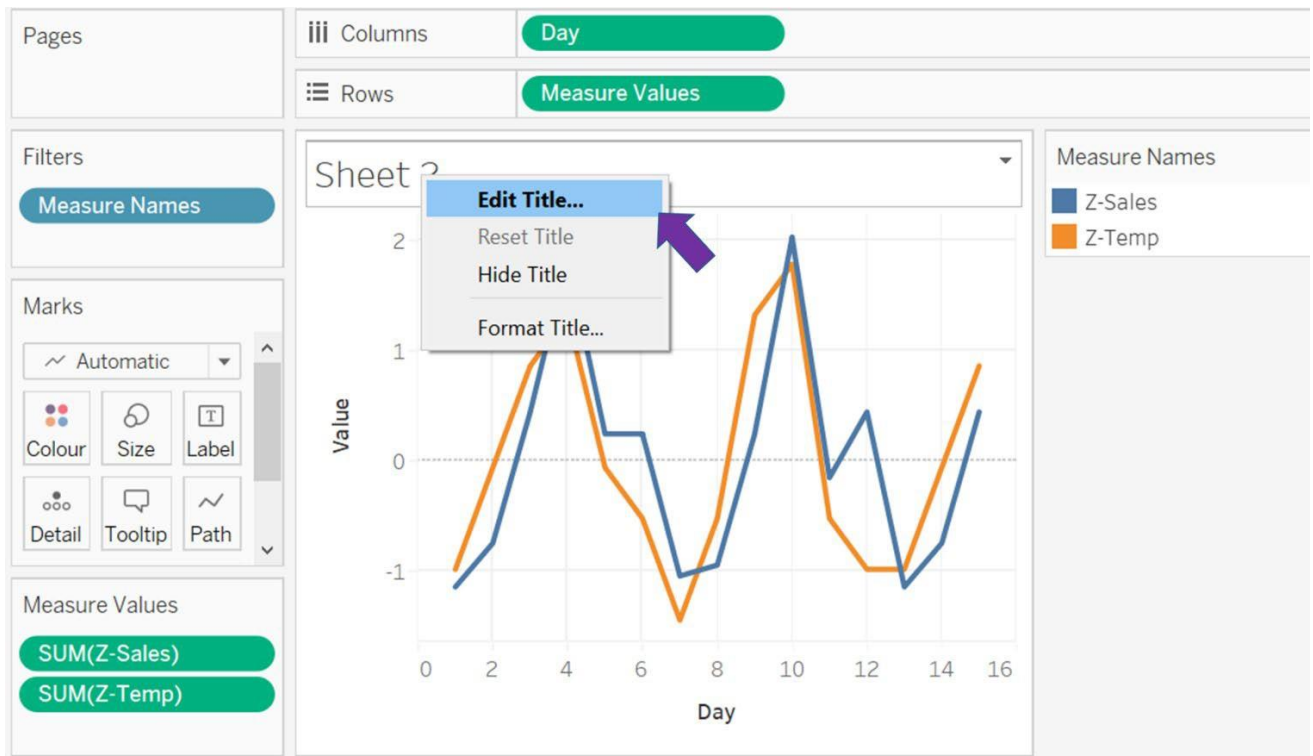
Now, you will be able to find which time of the day has the most submissions.

Appendix: How to create Title with your name

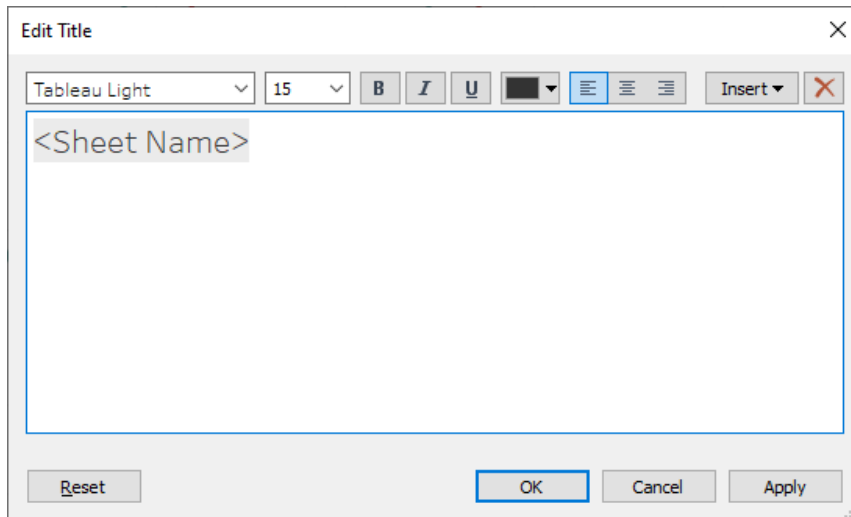
1. Original chart title follows the Sheet No.



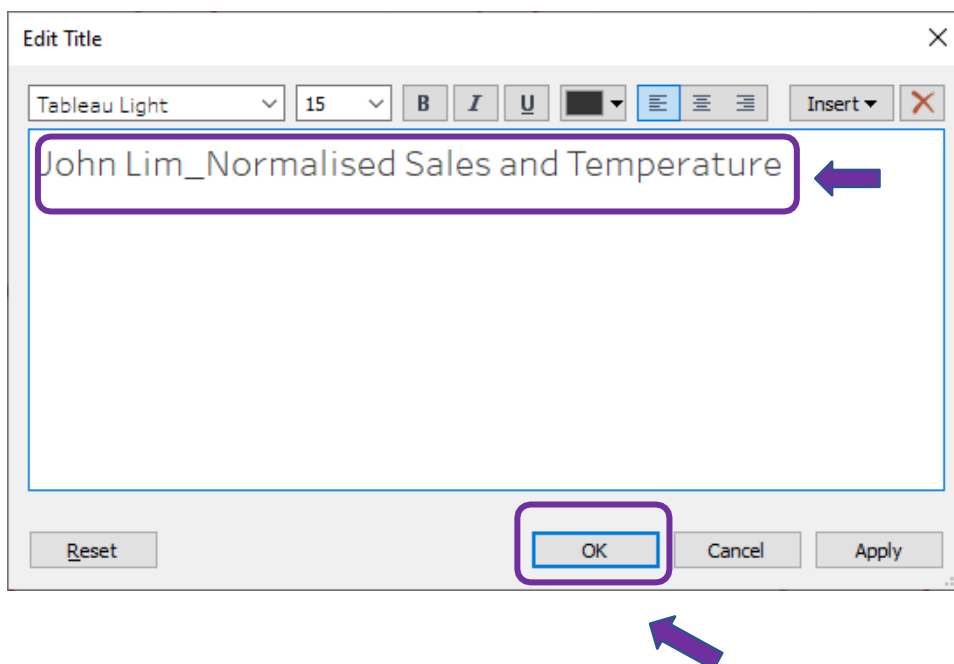
2. Right click on title -> Edit Title.



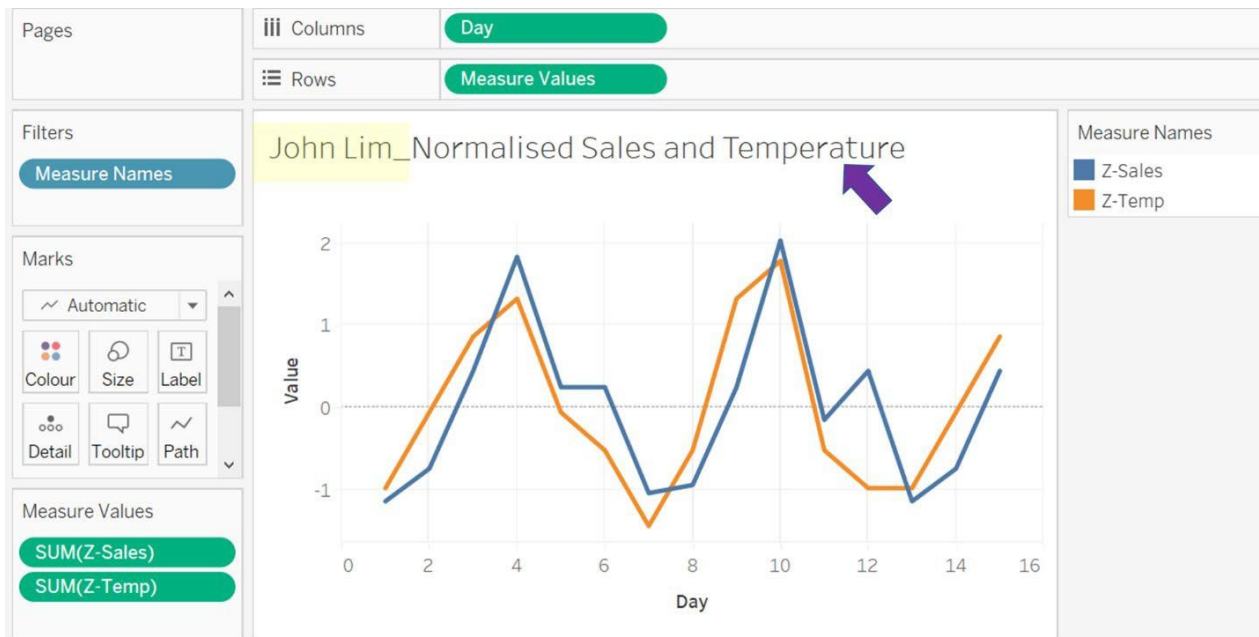
3. Edit Title pop up appears.



4. Now you can change the title accordingly. Make sure that you add your name for all submission especially for lab test (No marks will be awarded if the chart does not display your registered name for lab test). Click ok to complete the change.



5. Now the chart title contains your name and description.



Lab Exercise (optional)

On your own.

Create 2 calculated fields, named "X_Sales" and "X_Temp".

Apply the **Feature scaling** formula to transform the Sales and Temperature to X_Sales and X_Temp.

Plot the X_Sales and X_Temp on the same y-axis.